Engineering Standard

07 July 2024

SAES-X-300

Cathodic Protection of Marine Structures

Document Responsibility: Cathodic Protection Standards Committee

Table of Content

Summary of Changes

Paragraph Number		Change Type (Addition, Modification, Deletion, New)	Technical Change(s)
Previous Revision 02 May 2019	**Current Revision** 07 July 2024		
1.2	1.2	Addition	Boats and vessels added to Table-1
3.1	3.1	Addition	Changes as well as additional references included.
4.2	4.2	Addition	Changes as well as additional items added.
5.1	5.1	Modification	Note (2) modified to allow for multiple meetings.
5.2	5.2	Modification	Modified to include 90% detailed design package
5.4	5.4	Modification	Modified to be in line with other SAES and to include more details.
5.5	5.5	Modification / Addition	Modified to change proposal to detailed design. Detailed design calculations & surface area spreadsheets to be provided at 90% DD stage. Surface Area to be cross checked by Design Agency.
6.1.1	6.1.1	Modification	Modified to be in line with other SAES.
-	6.2.2	Addition	Addressed the overprotection and upper limit of CP potentials for different structures including carbon steel.
6.2.7/6.7.3	6.2.8/6.7.3	Modification	Simulation modeling requirements for complex structures specified.
6.3	6.3	Addition / Modification	Note (3) added. Note (2) is revised to clarify the requirement.
-	6.4.5	Addition	Note added for clarifications.
6.4.6	6.4.6	Modification	Changes made in Table 3 for current requirement calculations.

Paragraph Number		Change Type (Addition, Modification, Deletion, New)	Technical Change(s)
Previous Revision 02 May 2019	**Current Revision** 07 July 2024		
6.6.2	6.6.2	Modification	Reference of 6.2.2 added for maximum CP potential values.
6.7	6.7	Editorial	Section numbering revised to segregate requirement for Galvanic and Impressed Anode Design.
-	6.7.1.6	Addition	General requirements added to cover the bracelet anode thickness changes.
-	6.7.3	Addition	General design considerations added for addressing uniform current distribution.
6.7.5	6.7.1.5	Addition	Note 2 added in Table 4 for temperature and burial of pipeline anodes.
6.8.3 a	6.8.3 a	Modification	Revised to be in line with other SAES
6.9.6	6.9.6	Modification	Revised to limit DC voltage rating for T/R for pyramid anode and pile mounted anodes systems.
-	6.10.4	Addition	Voltage drop requirement for DC cables added.
6.11.4 a and b 6.11.7	-	Deleted	All requirements are sufficiently addressed in 6.11.5 & 17-SAMSS-012
6.13.1 a	6.13.1 a	Addition	Note added for bonding requirement to existing structure.
-	6.13.1 e	Addition	Requirement and clarification regarding bonding of support clamps.
7	7	Modification	Requirements added related to installation procedures and necessary liaison for Design Agency.

1 Scope

1.1 This standard prescribes the minimum mandatory requirements governing the design and installation of new cathodic protection systems for submerged new subsea pipelines, offshore platforms and other nearshore marine structures. The standard applies to both coated and uncoated structures.

1.2 This standard is applicable to but not limited to the marine structures categorized in Table 1

1.3 This standard is not applicable to corrosion protection of Splash Zone for any structure (see Section 6.4.5). This standard is not applicable to retrofits, modifications and repairs made to existing structures, but can be used as guideline for designing CP system, for existing structures. Moreover, any HDD installed pipeline with coating system other than CSD approved ARO system is also not covered by this standard.

2 Conflicts and Deviations

Any conflicts between this document and other applicable Mandatory Saudi Aramco Engineering Requirements (MSAERs) shall be addressed to the EK&RD Manager.

Any deviation from the requirements herein shall follow internal company procedure SAEP-302.

3 References

All referenced specifications, standards, codes, drawings, and similar material are considered part of this Engineering Standard to the extent specified, applying the latest version, unless otherwise stated.

3.1 Saudi Aramco References

Saudi Aramco Engineering Procedures

SAEP-302	Waiver of a Mandatory Saudi Aramco Engineering Requirement
SAEP-303	Engineering Reviews of Project Documentation
SAEP-332	Cathodic Protection Commissioning
SAEP-333	Cathodic Protection Monitoring

Saudi Aramco Materials System Specification

17-SAMSS-004	Conventional (Tap Adjustable) Transformer/Rectifiers for Cathodic Protection
17-SAMSS-006	Galvanic Anodes for Cathodic Protection
17-SAMSS-007	Impressed Current Anodes for Cathodic Protection
17-SAMSS-008	Junction Boxes for Cathodic Protection
17-SAMSS-012	Photovoltaic Power Supply for Cathodic Protection
17-SAMSS-017	Impressed Current Cathodic Protection Cables
17-SAMSS-018	Remote Monitoring Units (RMUs) for Cathodic Protection Applications

Saudi Aramco Engineering Standards

SAES-P-104	Wiring Methods and Materials
SAES-X-400	Cathodic Protection of Buried Pipelines
SAES-X-600	Cathodic Protection of External Surfaces of Plant Facilities

Saudi Aramco Best Practice

SABP-X-003	Cathodic Protection Installation Requirements

Saudi Aramco Standard Drawings

AA-036108	Offshore Negative Terminal Junction Box
AD-036132	Cathodic Protection Cable Identification Termination Detail
AA-036335	Cathodic Protection, Half Shell Bracelet Type Anode for Pipe Sizes 4-60 Inch
AA-036348	Cathodic Protection, Galvanic and Impressed Anodes on Offshore Structures
AA-036350	Cathodic Protection Bond Box-2 Terminal
AA-036384	Cathodic Protection Offshore Anode Junction Box
AA-036389	Galvanic Anodes Details
AB-036378	Cathodic Protection Onshore Rectifier Installation Details (Sheets 1 & 2)
AB-036907	Cathodic Protection, Test Stations for Buried Pipelines
AD-036785	Symbols for Cathodic Protection

Saudi Aramco Library Drawings

DA-950068-001	Bonding for Offshore Structures Installation Details

Saudi Aramco General Instructions

GI-0428.001	Cathodic Protection Responsibilities

3.2 Other References

International Organization for Standardization

ISO 15589-2, 2024	Petroleum, Petrochemical, and Natural Gas Industries - Cathodic Protection of Pipeline Transportation Systems - Part 2: Offshore Pipelines

Det Norske Veritas

RP-B401, 2017	Cathodic Protection Design
RP-F103, 2021	Cathodic protection of submarine pipelines

European Standard

EN 12473, 2024	General principles of cathodic protection in seawater

National Association of Corrosion Engineers

SP0176, 2022	Corrosion Control of Submerged Areas of Permanently Installed Steel Offshore Structures Associated with Petroleum Production

National Fire Protection Association

NFPA 70	National Electrical Code (NEC)

4 Terminology

4.1 Acronyms

GOSP	Gas and Oil Separation Plant
HAT	High Astronomical Tide Level
HDD	Horizontal Directional Drilled
ICCP	Impressed Current Cathodic Protection
LAT	Lowest Astronomical Tide Level
MIC	Microbiological Influenced Corrosion refers to corrosion mechanisms attributed to microorganisms and their by-products
MSAER	Mandatory Saudi Aramco Engineering Requirements
NEMA	National Electrical Manufacturers Association (USA
PMT	Project Management Team used as a truncated version of Saudi Aramco Project Management Team or SAPMT
RSA	Responsible Standardization Agent - usually the Saudi Aramco CSD cathodic protection Subject Matter Expert or the Supervisor of the CSD Cathodic Protection Team
SAES	Saudi Aramco Engineering Standard
SAMSS	Saudi Aramco Materials System Specification
SRB	Sulphate reducing bacteria, mainly responsible for MIC (Microbiological influenced corrosion) especially in anaerobic conditions
WIP	Water Injection Plant

4.2 Definitions

Anode: An impressed current or a galvanic anode for cathodic protection applications.

Anode Cable: A cable directly connected to an impressed current or galvanic anode.

Armored Cable: Armored cable for cathodic protection is a single core insulated cable manufactured with a double layer spiral wound insulated galvanized steel sheath to provide mechanical protection, and typically used for subsea applications.

Bond Cable: A cable installed between two metallic structures to provide electrical continuity between the structures for the purpose of cathodic protection.

Calcined Petroleum Coke Breeze: A carbonaceous backfill used as a conductive backfill media for impressed current anodes in soil.

Coated Casing: The term "coated casing" as used in this engineering standard describes a well casing with an external non-conductive coating (typically Fusion Bonded Epoxy or FBE). The coating must be applied to all sections of the casing in contact with soil or formation, from surface to the bottom of the casing or to a depth determined to facilitate external corrosion mitigation with cathodic protection through the relevant down hole corrosive aquifers. Casings that have been coated over the upper two or three joints of casing only are not "coated casings". Coating applied to well casings is not applied as a corrosion barrier. It is applied to reduce the total amount of CP required or to extend the influence of the applied CP.

CP Assessment Probe: A multi-electrode probe custom fabricated for Saudi Aramco that is designed to facilitate representative measurements of instant off, total polarization, current density, and resistivity.

CP Coupon: A cathodic protection coupon used to measure representative potentials or current densities on a pipeline or other buried or immersed metallic structure.

CP System Operating Circuit Resistance: The total resistance seen at the output of a CP power supply or the total working resistance in a galvanic anode system.

CP System Rated Circuit Resistance: The rated output voltage of a cathodic protection power supply divided by the rated output current. For photovoltaic power supplies, the rated output current is the design commissioning current.

Cross Country Pipeline: A pipeline between; two plant areas, another cross-country pipeline and a plant area, or between two cross country pipelines.

Deep Anode Bed: Anode or anodes connected to a common CP power supply and installed in a vertical hole with a depth exceeding 15 m (50 ft).

Design Agency: The organization responsible for the design of the pipeline and associated CP system. The Design Agency may be the Design Contractor, the Lump Sum Turn Key Contractor or an in-house design organization of Saudi Aramco.

Dielectric Shield This is an electrical insulator that is placed to avoid short between cathode and anode as electric charges do not flow through the material because they have no loosely bound, or free, electrons that may drift through the material.

Drain Point: The location on the cathodically protected structure where the negative cable from the T/R or negative junction box is fastened to the structure.

Electrified / Energized Structure: In terms of CP, all offshore structures especially platforms where either enough AC power currently available or in the immediate future (within 3 years from installation to up to maximum of 10 years as per project requirements) for the operation of ICCP system.

Foundry (for anodes): An anode foundry is a facility that produces metal castings from either ferrous or non-ferrous metals or alloys.

Galvanic Anodes: Anodes fabricated from materials such as aluminum, magnesium or zinc that are connected directly to the buried structure to provide cathodic protection current without the requirement for an external cathodic protection power supply. Galvanic anodes are also referred to as sacrificial anodes.

Hazardous Areas: An area where fire or explosion hazards may exist due to flammable gases or vapors, flammable liquids, combustible dust, or ignitable fibers or filings (see NEC Article 500).

HISC: Hydrogen-induced stress cracking is a phenomenon caused mainly by hydrogen embrittlement in the presence of load. Cause of hydrogen embrittlement is associated with the ingress of hydrogen at steel surface due to cathodic polarization.

Impressed Current Anodes: Anodes typically fabricated from High Silicon Cast Iron (HSCI), Mixed Metal Oxide (MMO) or liner anode, etc. that are connected through a DC power supply to the buried or immersed structure to provide cathodic protection current.

Manufacturer: The Company that assembles the components for the finished product and provides the finished product either through a Vendor or directly to Saudi Aramco.

Mudline: The ocean floor at the location of interest.

Nearshore Marine Structures: For the scope of this document (Other) nearshore marine structures are all offshore structures other than platforms and subsea pipelines.

Negative Cable: A cable that is electrically connected (directly or indirectly) to the negative output terminal of a cathodic protection power supply or to a galvanic anode. This includes bond cables to a cathodically protected structure.

Non-electrified / Non-energized Structure: In terms of CP, all offshore structures other than electrified.

Off-Plot: Off-plot refers to any area outside of the plot limits.

On-Plot: On-plot refers to any area inside the plot limit.

Perimeter Fence: The fence which completely surrounds an area designated by Saudi Aramco for a distinct function.

Photovoltaic Module: A number of solar cells wired and sealed into an environmentally protected assembly.

Pipeline: The term "pipeline" is used generically to refer to any type of pipeline.

Plant Area: The area within the plot limits of a process or storage facility.

Platform: An offshore structure used to accommodate oil and gas wells, related production equipment, pipelines, and living quarter.

Plot Limit: The plot limit is the boundary around a plant or process facility. The plot limit may be physical such as a fence, a wall, the edge of a road or pipe rack, chains and posts or a boundary indicated on an approved plot plan.

Positive Cable: A cable that is electrically connected (directly or indirectly) to the positive output terminal of an impressed current cathodic protection power supply, including impressed current anode cables.

Process Pipeline: A pipeline typically associated with a plant process and typically above ground within a plant facility.

Production Pipeline: A pipeline transporting oil, gas or water to or from a well. These include flowlines, test lines, water injection lines, and trunk lines.

Pyramid Anode: An offshore anode assembly custom manufactured for Saudi Aramco that is fabricated with mixed metal oxide anode components and

	placed on the sea bed. The pyramid anode derives its name from the pyramid shaped concrete base.
Resistivity Meter:	A four terminal meter designed to measure ground resistivity, or can be connected to measure resistance in a format that excludes the resistance of the test wires.
Reference Electrode:	An industry standardized electrode used as a common reference potential for cathodic protection measurements. A copper/copper sulfate (Cu/CuSO4) reference electrode is typically used for soil applications. A silver/silver chloride (Ag/AgCl/0.6M Cl) reference electrode is typically used for aqueous applications.
Soil Transition Point:	The surface location where a pipeline enters or exits the soil, i.e., above grade to below grade transition, or below grade to above grade transition.
Splash Zone (for CP):	Splash Zone is defined as the zone of an offshore structure that is alternatively in and out of the water because of waves, tides, winds and sea. For details see Section 3.2 of NACE SP0176-2007.
Stray Current:	Current flowing through the unintended circuits/paths.
Subject Matter Expert (SME):	For the purposes of this standard, the SME shall be the assigned Consulting Services Department cathodic protection specialist.
Submerged Zone:	Surface Area of the marine structure that is always covered with water and extends downwards from the splash zone and includes that portion of the structure below the mudline.
Subsea Pipeline:	A submarine pipeline (also known as marine, subsea or offshore pipeline) is a pipeline that is laid on the seabed or below it inside a trench. In some cases, the pipeline is mostly on-land but in places it crosses water expanses, such as small seas, straights, and rivers.
Surface Anode Bed:	Anode or anodes connected to a common CP power supply, installed either vertically or horizontally at a depth of less than 15 m (50 ft).
Tension String Anode:	An offshore anode assembly custom manufactured for Saudi Aramco that is fabricated with mixed metal oxide anode components and suspended on a wire rope/tension spring assembly beneath an offshore platform.
Test Line:	A pipeline that is used for testing an individual well or group of wells.
Transformer/Rectifier (T/R):	A CP power supply that transforms and converts AC to DC power, often referred to in the cathodic protection industry as a rectifier.
Transmission Pipeline:	A cross country pipeline transporting product between GOSPs WIPs or other process facilities.
Trunk-line:	A pipeline designed to distribute or gather product from two or more wells, typically connecting flowlines or injection lines to the associated GOSP or WIP.
Uncoated Casing:	The term “uncoated casing” when used in this engineering standard describes a well casing that is either bare (no external coating), or may have an external coating applied to the shallow sections of casing to minimize corrosion in the landing base or surface soils.

Utilization Factor:	Utilization factor is a constant value, depending upon the shape of anode and how it is attached, relates to reduction in anode size due to consumption to the minimum size before it ceases to provide the required current output.
Utility line:	A pipeline that delivers a service product (typically water, gas or air).
Wellhead Platform (WHP):	The platform located above the head of a gas or oil well and use to drill and operate the well.

5 Design Review and Approval

5.1 The proposed construction drawings and the related cathodic protection design information for every design package shall be submitted to the CP Proponent organization (as defined by GI-0428.001) and to Saudi Aramco's Consulting Services Department (CSD) for review and approval.

Notes:

(1) As noted in GI-0428.001, PMT may request assistance from CSD for the verification of the qualifications of the Design Contractor's engineer responsible for designing the CP systems.

(2) It is recommended that the CP Design Agency hold meeting(s) but not limited to a pre-design meeting with CSD and the Proponent CP Group to outline and agree to the design approach for the Project Proposal and Detailed Design.

5.2 The Design Agency shall not issue drawings for construction until the 90% detailed design package including drawings have been reviewed and approved in writing by CSD and the CP Proponent organization.

5.3 Product hardware and software specifications and relevant design information for all remote monitoring equipment proposed for cathodic protection systems shall be submitted to the Supervisor of IT/COD/SCSD (Information Technology/Communication Operations Dept./Satellite & Communication Support Div.) for review and approval during the Project Proposal and Detail Design phases of the Project.

5.4 The Project Proposal package shall be submitted to CSD's review if required by PMT. The Project Proposal package shall provide all general design considerations that can be developed without requiring measurement of field data. The field data including existing CP system details were available can be provided as an input by the PMT team by collecting the same from Proponent. Cathodic protection requirements shall be based on the following consideration which shall be clearly specified and agreed with the Proponent at the project proposal stage.

a. the new structure and associated CP equipment to be an independent system.
b. the new structure as a part of the existing system which considers spare capacity of existing system, existing protection status and upgradation of existing system

The Project Proposal package shall include:

a. A scope of work including a specific statement that clearly identifies any additional requirement to provide CP for any existing structure
b. Proposed locations of new CP system(s) on an overall CP system layout drawing, including proposed anode type(s) and estimated output ratings of the proposed cathodic protection power source (in case of ICCP).
c. Information on existing cathodic protection system location and spare capacity (from nearby or related system).
d. Information on all locations where the proposed structure will be mechanically connected to other existing structure with clear details of bonding/isolation.

e. Proposed anode distribution and installation details.

5.5 The 90% detailed design package shall be submitted to the CP Proponent organization and to Saudi Aramco's Consulting Services Department (CSD) for review and approval.

The detailed design package shall include all detailed information and documents as per Table-1 of SAEP-303 and not limited to following:

a. The detailed scope of work / basis of the CP design with reference to project specifications and applicable standards.
b. List all Critical CP Design's assumptions, limitations and exclusions clearly.
c. Calculations of surface area of the structure to be protected.

Notes:

(1) Surface area calculations shall be verified by PMT and/or proponent, cross checked by Design Agency, review and approval of any design packages by CSD is based on verified surface area values provided in the design package.

(2) Any changes to the surface area after CSD approval of CP design package shall be included in the revised design CP Design package and this revised design package shall be submitted for review & approval by CSD.

d. All design calculations and applicable field data required to verify design compliance with Saudi Aramco Cathodic Protection Engineering Standards.
e. Detailed design calculation spreadsheets (editable) including surface area calculation spreadsheets.
f. Professionally drafted full size Index "X" CP drawings that:

- Detail each CP item by description and stock number if applicable
- Detail the proposed location for each piece of CP equipment including but not limited to T/Rs, anodes, junction boxes and test stations
- Galvanic anode installation details
- Detail and specifically identify all cathodic protection cables including all anode, structure, bond, and T/R cables
- Clearly identify the specific and individual cable routing and termination points within the respective junction boxes, bond boxes, and T/Rs
- Detail all cathodic protection equipment using the cathodic protection symbols shown on Standard Drawing AD-036785 "Symbols for Cathodic Protection"
- All details for the Remote Monitoring System including details on the hardware, software, and connectivity

6 General Design Requirements

6.1 Contractor/Designer Qualifications

6.1.1 Cathodic protection designs shall be carried out by design offices that are approved by Saudi Aramco. Such design offices must have qualified engineers with a minimum of ten years verifiable cathodic protection design experience and a minimum industry qualification of NACE CP Level 4 and BS degree. Indicate the NACE CP level, certification number, name of designer and/or his signature/initial in the design package.

Note:

For CP designs related to categories A and B of Table 1, design experience shall be related to relevant structures.

6.1.2 Field measurements required for the design shall be collected by an Engineer or Technician with a minimum of five years verifiable cathodic protection field experience and industry certification level of NACE CP Level 2 or higher. Installation supervision of cathodic protection materials shall be done by Engineer or Technician with a minimum of five years verifiable cathodic protection field experience and industry certification level of NACE CP Level 2 or higher. All design calculations and applicable field data required to verify design compliance with Saudi Aramco Cathodic Protection Engineering Standards.

6.2 General

6.2.1 For "Limitations of CP" and "Detrimental effects of CP" see sections 5.2 and 5.5 of DNV-RP-B401 (2017). Also see Section 4.6 "Precautionary Notes" of NACE SP0176. CP Designer (Design Agency) shall consider all relevant limitations and detrimental effect of CP while designing a CP system and include the same in CP design report.

6.2.2 To avoid overprotection and any risk of HISC for structure to be protected, maximum negative potential mentioned in Table 2 of section 6.2 of EN12473 shall be applicable and for other materials section 6.3 shall also be applicable as appropriate. For duplex or martensitic stainless steel as these alloys can be susceptible to HISC and hence potentials more negative than -0.80V (as per ISO 15589-2) shall be avoided.

6.2.3 CP system shall be designed to provide**:**

- Cathodic protection for all submerged and buried metallic marine structures, equipment and pipelines using galvanic anode systems or impressed current anode systems (with galvanic anode as temporary CP).
- Effective current distribution so that the protection criteria is effectively achieved on the entire surface.
- Design life of CP system equal to or more than the required design life.
- Monitoring facilities to evaluate system's performance.
- Enough allowance for anticipated changes in requirements with time and environmental conditions.

6.2.4 CP system shall be designed considering the effect of environmental conditions and any adjacent neighboring structure. Proposed design shall be compatible with existing CP system of all nearby pipeline and structure to avoid current drain from one system to another.

6.2.5 Design the cathodic protection system to include the equipment that is required for monitoring the protected structures per SAEP-333. Install the equipment in accessible unobstructed locations.

6.2.6 Cathodically protect both submerged sections and buried sections of subsea pipelines that terminate inside onshore facilities, or up to the isolation flange for the onshore-offshore transition.

6.2.7 Protect the submerged sections of subsea pipelines with bracelet galvanic anodes according to Standard Drawing AA-036335. Details of different types of galvanic anodes are provided in the drawing AA-036389. For subsea pipelines, bolted anodes will be required unless bolting arrangement is not possible due to installation limitations, then "welded anodes" can be used instead. For onshore sections of subsea pipelines, refer to SAES-X-400 or SAES-X-600, as applicable.

6.2.8 For following complex scenarios, CSD shall be involved right from early stages including pre-design/conceptual stage. Simulation modeling for CP designs shall be carried out in addition to specialized monitoring surveys as additional design/commissioning approval requirements:

a. Extension of existing marine structures.
b. Designs with hybrid CP Systems for permanent CP (Both Galvanic and ICCP system) or two different types of anodes (different anode materials or different anode mounting, etc.) for

same structure.

Exception:

Subsea structures are usually protected considering Temporary CP system with Galvanic Anodes and Permanent CP system with ICCP system, this shall not be considered as a requirement for simulation modelling.

c. Designs involving HDD pipelines regardless of the length of pipeline.
d. Subsea pipelines with ICCP systems.

6.2.9 For offshore platforms with well casings, an impressed current cathodic protection (ICCP) system shall be provided where AC power is available (Electrified / Energized Structures).

6.2.10 Galvanic & Impressed Current Anodes arrangements for offshore structures are provided in reference drawing AA-036348 and for Offshore Anode Junction Box refer to drawing AA-036384.

6.3 Design Life

Minimum design life for the cathodic protection systems shall be 25 years as follows:

a. Galvanic anodes only: 25 years (preferably for offshore pipelines and non-electrified / non-energized structures) and
b. Impressed current anodes: 25 years (preferably for electrified / energized structures)

Notes:

(1) In addition to 25 years of permanent design life, ICCP system shall include galvanic anodes CP for minimum of 3 years (and maximum of 10 years as per project requirements) as temporary CP from installation / construction period and/or till ICCP system is turned ON. For further requirements regarding temporary CP see Note 6 of Table 2.

(2) If for any reason the time for energization (ICCP ON) is more than 10 years or unknown, then the structure shall be considered as non-electrified / non-energized and CP system shall be designed accordingly.

(3) When permanent SACP system is designed for any structure, it shall be provided within 90 days to all sections of structure which become submerged. If for any reason permanent SACP cannot be installed within this period, temporary CP shall be provided till permanent SACP become functional. This requirement is applicable for speific structures such as sheet piles in Category C where construction activity commences in batches and not applicable to other structures (category A, B & C of Table 1), as these structures are provided with pre-installed anodes before site construction acitivites.

6.4 Current Density Criteria

6.4.1 The CP system design shall provide the minimum current detailed in Table 2 and Table 3.

6.4.2 Impressed Current CP systems for offshore platforms shall provide sufficient current for all submerged surface areas, including the total submerged and buried surface area of the piles, well casings, conductors, plus all submerged and buried sections of other structures associated with the platforms.

6.4.3 The design for the cathodic protection system for electrified platforms with well casings shall provide the currents listed in Table 2 for each well casing in addition to the current required for the surfaces of the platform structure that are submerged and below the mud-line.

6.4.4 Galvanic Anode CP systems for offshore platforms shall provide adequate current for all submerged surface areas, plus:

- The current required for each casing as specified in Table 2,

- The buried surface area of the piles and conductors, and
- All submerged and buried sections of other structures associated with the platform.

6.4.5 The submerged structure surface area is defined as the total area extending from the base of the structure to Lowest Astronomical Tide (LAT) level. For CP calculations, anode requirement shall be calculated and considered from the base of structure to High Astronomical Tide (HAT) level (including splash zone). However, cathodic protection is not effective in the splash zone, corrosion protection in this zone depend on the high quality coating and other appropriate means.

Note:

Surface Area of any metallic sleeve (i.e. Monel sheathing) shall be considered as bare structure for calculation of current requirement / anode quantity.

6.4.6 For offshore pipelines, quantity and spacing of bracelet anode requirements as specified in Table 3 are applicable at ambient temperature and for fluid internal temperature up to 50°C, it shall be adjusted for fluid temperature more than 50°C.

6.4.7 For offshore pipelines, if for any reason the maximum distance between the anodes mentioned (150 m) in the Standard Drawing AA-036335 is to be increased, then it shall be subject to proponent's approval and based on attenuation calculations as per Annex B of ISO 15589-2:2012. Also refer to Section 6.2.8 for HDD pipelines.

6.5 Resistivity

6.5.1 For calculation purposes, the resistivity of seawater shall be assumed to be 17 ohm-cm unless measurements taken at the design location demonstrate a more accurate value.

6.5.2 For anode output calculations (resistance) especially in case of anodes laid on seabed a more conservative resistivity (51 ohm-cm) shall be considered depending upon the location and estimating self-burial.

6.6 Protection Criteria

6.6.1 The cathodic protection system shall achieve a minimum structure-to-electrolyte potential of negative 0.90 volt with reference to a silver/silver chloride electrode. This is also applicable for buried anaerobic environments and where possibility of SRB activity exist.

6.6.2 Maximum limit for structure-to-electrolyte potential shall be considered in as per Section 6.2.1 and 6.2.2

6.6.3 For onshore portions of marine structures, it is required to achieve negative structure-to-soil potentials in compliance with SAES-X-400 and SAES-X-600.

6.7 Anode Design

6.7.1 Galvanic Anodes

6.7.1.1 Anode resistance shall be calculated using equations based on the shape of anode selected as per Section A-8 in Annex A of ISO 15889-2 and Table A-7 of DNV-RP-B401.

6.7.1.2 Anode utilization factor shall also be selected based on the shape of anode from section 8.4 of ISO 15889-2 and Table A-8 of DNV-RP-B401.

6.7.1.3 Final required anode quantity shall be calculated considering section 7.9.1 to 7.9.6 of DNV-RP-B401 and other design inputs and results shall also be considered from simulation modeling etc. (see section 6.2.8)

6.7.1.4 For anode design, section 7.10.3 of DNV-RP-B401 and for distribution of anodes, section 7.11 of DNV-RP-B401 shall be considered.

6.7.1.5 Use galvanic anodes that comply with the appropriate sections of 17-SAMSS-006. Table 4 shows the recommended design parameters for galvanic anode materials under normal condition.

6.7.1.6 For offshore pipelines, if for any reason the minimum thickness of the anodes mentioned in the Standard Drawing AA-036335 is to be decreased, then this shall be limited to 40 mm. To support this, the design agency shall provide justification and alternative design calculations in the detailed design justifying that the alternative anode dimensions shall:

a. Meet minimum anode weight required and individual anode current output & design life.
b. Demonstrate with calculations that reduced thickness will not result in accelerated anode consumption due to localized corrosion and/or higher utilization, etc.

6.7.2 Impressed Current Anodes

6.7.2.1 Use one of the following impressed current anode materials, according to specification 17-SAMSS-007:

a. Mixed Metal Oxide (MMO)
b. Platinized Niobium (Pt/Nb)
c. High Silicon Cast Iron (HSCI) for shoreline area only

6.7.2.2 Refer to Table 5 for the recommended design parameters for the approved impressed current anode materials.

6.7.2.3 If various types of impressed current anode material are used on a structure, connect only anodes of the same material to any one T/R.

6.7.3 Regardless of the type of anodes (galvanic or impressed), the anodes shall be distributed to provide uniform distribution throughout the structure. Where simulation modelling is considered (see section 6.2.8) the anode distribution shall follow the simulation modelling results. For others following parameters shall be considered in design:

a. Geometry of the structure.
b. Vertical and horizontal structure members.
c. Anode and structure surface area & correlation between structure current requirement and anode current output.
d. Good balance between anode quantity based on current and based on weight for Galvanic Anode systems

Note:

To achive proper current distribution more number of smaller weight anodes instead of less number of larger weight anodes shall be considered. This approach shall be considered especially while designing the temporary CP system.

6.8 Circuit Resistance

6.8.1 For a T/R, the CP system rated circuit resistance shall be defined as the T/R rated voltage, divided by the T/R rated current. Rated voltages and currents are as detailed on the manufacturer's data sheet/plate.

6.8.2 The CP system operating circuit resistance for an ICCP system shall be defined as the total effective resistance seen by the output terminals of the respective ICCP power supply, and for calculation purposes shall include:

a. Anode resistance to water.
b. Positive cable resistance from CP power source to anodes.
c. Negative (cable)s resistance from CP power source to structure.
d. Effective resistance caused by +0.95 volts anode back emf
(for MMO in sea water plus the structure back emf -1.0 volts (example: -1.0 volts casing

back emf + 0.95 volts anode bed back
emf = 1.95 volts total between the anodes and structure).

6.8.3 ICCP system designs shall take into consideration the calculated operating resistance and shall size the positive and negative cables and voltage rating of the T/R such that the "calculated" operating output of the T/R complies with all of the following:

a. At the design stage only, the target commissioning current shall be achieved at a voltage of lower than 70% of the T/R rated voltage output.

b. The normal operating output current shall be achieved with the voltage adjustment set at more than 10% of the available (rated) T/R voltage output.

Note:

The above section is a mandatory design requirement but is not a mandatory requirement for commissioning acceptance.

6.8.4 The CP system "operating" circuit resistance measured during commissioning of a new CP system shall not be greater than 90% of the CP power supply "rated" circuit resistance.

6.9 DC Power Source

6.9.1 ICCP systems shall be designed with CP power supplies sized with a minimum rated output current equaling the sum of the current requirement plus a design surplus.

Note:

The design surplus is required to ensure that the ICCP power supplies will not be operated at full rated current capacity. In most of the cases the spare current is considered in the design and standard rating of the rectifier will be selected which is more than meeting the required current and hence has built-in design surplus. However, in rare cases where the standard T/R rating does not meet the design surplus, then additional allowances to the T/R current rating shall be added.

6.9.2 Cathodic protection rectifiers and photovoltaic systems shall be manufactured in accordance with 17-SAMSS-004 and 17-SAMSS-012 respectively and shall be equipped with remote monitoring equipment in accordance with17-SAMSS-018.

6.9.3 Install the CP power supplies in accordance with Saudi Aramco Library Drawing DA-950035 sheet 1, or Saudi Aramco Standard Drawing AB-036378 sheets 1 and 2.

6.9.4 Use oil immersed T/Rs (Type ONAN) inside hydrocarbon plant areas, for all outdoor marine environment applications, and for Class I Division 2 (Zone 2) applications.

6.9.5 Use air cooled T/Rs (Type AA) for indoor or non-classified, non-marine environment applications. T/Rs with NEMA Class 3R enclosures shall NOT be used inside hydrocarbon plant areas or in marine environments unless installed indoors.

6.9.6 DC power supplies shall have a maximum rated output voltage of no greater than 50 volts for offshore including landfill areas. The sizing of the T/R shall be optimized based on the circuit resistance in accordance with Section 6.8 of this standard.

Note:

It is noted that there are some existing pile mounted anodes and in such cases 30 volts limit is to be applied.

6.9.7 For hazardous areas (maximum Class I, Zone 2), the design agency shall select a cathodic protection DC power supply (and other CP system equipment) that complies with the requirements of NEC Articles 500 to 504 for hazardous (Class I, Zone 2) areas. CP equipment shall not be placed in Class 1 Zone 1 areas.

6.10 DC Cables

6.10.1 DC cables connected to the T/R output terminals shall comply with 17-SAMSS-017 and shall be:

a. 25 mm^2 or larger, and

b. Sized in compliance with the National Fire Protection Association NFPA 70, National Electrical Code (NEC).

Note:

For typical applications, use the 90°C column in Table 310.15(16) and refer to 310.15(B)(2) to consider an ambient temperature of 40°C to calculate the ampacity for HMWPE cables.

6.10.2 Cable terminations shall comply with Standard Drawing AD-036132.

6.10.3 Platforms with impressed current CP shall have a dedicated negative cable to each well casing. Use a negative junction box (AA-036108) to terminate multiple negative leads. Junction and bond boxes used in cathodic protection systems for marine structures shall comply with the design and construction required by 17-SAMSS-008.

6.10.4 Size and length of cables (if considered) in galvanic anode CP systems designs shall be selected to limit DC voltage drop in cables not more than 10% of the driving voltage.

6.11 Monitoring Facilities

6.11.1 For the onshore portions of marine pipelines, provide test stations for measuring pipe-to-soil potentials at insulated cased crossings, paved road crossings and other locations outside the SSD fence and plant fence as required by operational needs. Use Standard Drawing AB-036907 as a reference for the design and construction of the test station.

6.11.2 For special cases where CP monitoring of offshore pipelines are restricted such as longer lengths of HDD pipelines, design and commissioning of CP system shall include additional measures which may include but not limited to attenuation calculations, simulation modeling and specialized monitoring surveys etc.

6.11.3 Where a remote monitoring system (RMS) is used, the design details for the RMS shall be submitted to CSD and the CP Proponent organization for review and approval. Mandatory design details include:

a. A description of the existing CP RMS used by the respective CP Proponent in the respective area (if applicable). The description shall detail existing RMS hardware, software, communication connectivity and protocols.

b. Confirmation of proposed hardware and software compatibility with the existing CP RMS (if applicable), or written authorization by CSD and the CP Proponent organization to use a non-compatible RMS.

c. A data flowchart that illustrates the data path from the CP power supply to the CP Proponent's desktop including connectivity details with software requirements and communication protocols.

6.11.4 Remote monitoring systems for offshore cathodic protection power supplies shall be designed (preferably with provision of remotely controlled) and installed at all new ICCP system installations.

6.11.5 Remote Monitoring Units (RMUs) for cathodic protection power supplies shall comply with the requirements of 17-SAMSS-018.

6.11.6 Cables and field wiring used for the remote monitoring systems shall comply with SAES-P-104.

6.11.7 Test posts remote monitoring units shall be provided for the sub-sea pipelines in the onshore to offshore transition locations with provision to monitor both the onshore and offshore side

6.12 Junction and Bond Boxes

Cathodic Protection Junction Boxes and Bond Boxes shall be manufactured in accordance with 17-SAMSS-008.

6.13 Bonding

6.13.1 Bonding shall be installed to ensure complete electrical continuity throughout a marine structure.

a. In case of galvanic anode CP system, all bolted or clamped components with a surface area exceeding 1 m^2 shall have resistance less than 0.1 ohm otherwise such components shall have an all-welded or brazed connection to an anode. The minimum size for cables shall be 16 mm2 (6 AWG). At least two cables per anode shall be installed.

Note:

Electrical continuity with existing structure can be achieved by a mechanical connection using e.g. serrated washers to provide a reliable electrical connection at bolt heads or washers as long as the resistance requirement of less than 0.1 ohm can be achieved and demonstrated.

b. For ICCP systems the minimum size for cathodic protection bond cables shall be 70 mm^2 (2/0 AWG).

c. Install steel bond bars or bond cables to ensure electrical continuity between each well casing, its associated conductor/jacket, and the platform, in accordance with Library Drawing DA-950068-001.

d. Install bond cables for J-tubes, boat landings, riser guards or other submerged mechanical fittings that are not welded to the primary marine structure.

e. All support structures such as clamps for J Tube, Risers and / or Caissons as well as all non-welded structures shall be bonded to parent structure. This shall be documented clearly in design document and drawings.

6.13.2 Use flexible straps or bond cables at both ends of the walkway to electrically bond walkways connecting two adjacent platforms or other type of marine support structures. Provide at least two bonds at each end.

6.14 Electrical Isolation

6.14.1 Do not use isolating devices on offshore structures.

6.14.2 Do not use isolating devices on submerged or buried portions of pipelines.

6.14.3 Provide aboveground electrical isolating devices to isolate pipelines at the onshore-offshore transition.

6.14.4 Install a 2-terminal bond box at each isolating device in an accessible location. The bond box design and construction shall comply with Standard Drawing AA-036350.

7 Installation, Records, Commissioning, and Inspection

Refer to Saudi Aramco Best Practice SABP-X-003. SABP-X-003 shall be deemed a mandatory document for this Standard.

In addition to SABP-X-003, refer to Section 7 “Control of Interference Currents” and Section 8 “Dielectric Shields” of NACE SP0176.

As built drawings and documents including installation procedures and Commissioning Reports shall be submitted to Proponent for review and approval. Design Agency shall liaison with installation and commissioning contractor till the successful commissioning of the CP system.

Commissioning Reports shall be submitted to CSD for information and feedback.

Refer to Saudi Aramco Engineering Procedures SAEP-332 for commissioning and SAEP-333 for monitoring procedures.

Table 1: Categories of Marine Structures

A) Subsea Pipeline	B) Offshore Platforms	C) Other Nearshore Marine Structures
Pipeline laid on seabed Open trench/buried pipeline HDD Installed Pipeline	Pumping Platforms Auxiliary Platforms Utility Platforms Metering Platforms Tie-in Platforms Gas/Oil Separator Platforms (GOSP) Gas Compression Platforms Well Platforms (including jackets and casings) Living Quarter Platforms	Mooring Dolphins Sea Islands Breasting Dolphins Trestles Flare Support Structures Break Waters Pipe Supports Pipeline Markers Loading/Mooring Buoys Piers Piling (including sheet piling) Wharves Single Point Mooring (SPM) Boats and vessels.

Note:
For subsea pipeline, this standard shall be the only governing design standard for CP system, if pipeline is electrically continuous from the Onshore/Offshore transition to its other end. SAES-X-400/SAES-X-600 shall be considered only for onshore sections of pipeline.

Table 2 - Design(1) Current Requirement for Offshore Structures (Categories B & C only)

Offshore Structure	Type of Cathodic Protection	Coated(2) & (3)		Bare(2) & (3)	
		Seawater	Mud/Soil(7)	Seawater	Mud/Soil(7)
Platform(2) and (3) (Current for platform surfaces)	ICCP	13 mA/m²	4 mA/m²	65 mA/m²	20 mA/m²
	Galvanic only(4)	15 mA/m²	6 mA/m²	75 mA/m²	30 mA/m²
Oil(2) Well Casing (Current(6) for each well)	ICCP or Galvanic	10 amps		35 amps	
Gas(3) Well Casing (Current(6) for each well)	ICCP or Galvanic	50 amps		Not applicable	
Other	ICCP or Galvanic	13 mA/m²	4 mA/m²	65 mA/m²	20 mA/m²

Notes:

(1) Total current requirement calculated using Table 2 are for design purposes to determine the number of galvanic anodes and the minimum rated current capacity of the impressed current CP power supply where applicable.

(2) Electrified / Energized Oil Fields structures with well casings (including water injection wells), do not need to be coated (structure as well as well casing), however ICCP system shall be designed to provide CP. Electrified / Energized Oil Fields without well casings, can have either coated structure with galvanic CP or bare structure with ICCP system. Non-electrified / Non-energized Oil Fields with or without well casings, shall have coated structure and coated well casings up to 4,000 ft. with galvanic CP. This is further clarified in Table form attached as Appendix 1.

(3) Electrified / Energized Gas Fields with well casings, shall have coated structure and coated well casings up to 10000ft. ICCP system shall be designed to provide CP. Electrified / Energized Gas Fields without well casings, can have either coated structure with galvanic CP or bare structure with ICCP system. Non-electrified / Non-energized Gas Fields with or without well casings, shall have coated structure and coated well casings up to 10,000 ft. with galvanic CP. This is further clarified in Table form attached as Appendix 1.

(4) Current density requirements for a "galvanic anode only" cathodic protection system for "platforms with well casings" is higher than "ICCP", because ICCP systems use negative drain points connected directly to the well casings, whereas "galvanic only" are not connected immediately to the well casings and therefore experience additional comparative attenuation before protecting the well casings. For platforms with no well casing, current density values of "ICCP" shall be used.

(5) Cathodic protection systems for pipelines or electrically isolated sections of a pipeline that require 2.0 amperes or less may use galvanic anodes.

(6) A current drain of 5 Amps per well casing shall be considered for temporary CP system. However, current density values will be same for permanent and temporary CP system design for rest of the structure.

(7) All buried structures such as piles and bottom of Mudmats (in contact with seabed) etc. shall be considered bare (even if coated) for calculating current requirements where significant damage to paint coatings during fabrication and installation is anticipated.

Table 3 - Design Current Requirement for Offshore Pipelines (Category A only)

<table>
<tr><th rowspan="2">Offshore Structure</th><th rowspan="2">Type of Cathodic Protection</th><th colspan="2">Coated</th><th colspan="2">Bare</th></tr>
<tr><th>Seawater</th><th>Mud/Soil</th><th>Seawater</th><th>Mud/Soil</th></tr>
<tr><td>Pipeline Laid on Seabed (with concrete coating)</td><td>Bracelet Anodes</td><td rowspan="2" colspan="4">See Standard Drawing AA-036335 for minimum bracelet galvanic anode requirements. A verification design calculation shall be submitted to confirm that these anode quantities are adequate using the current density, coating break down factors values from DNV-RP-F103 and anode quantity increased if required by the calculations.</td></tr>
<tr><td>Pipeline Laid on Seabed (without concrete coating)</td><td>Bracelet Anodes</td></tr>
<tr><td>Buried(1) Pipeline or installed by open cut trench method</td><td>ICCP, Galvanic(2), or ICCP plus Galvanic</td><td colspan="2">4 mA/M^2</td><td colspan="2">Not Applicable</td></tr>
<tr><td>HDD(3) installed Pipeline (CSD approved ARO coating system)</td><td>ICCP, Galvanic(2), or ICCP plus Galvanic</td><td colspan="2">6 mA/M^2</td><td colspan="2">Not Applicable</td></tr>
<tr><td>Other</td><td>ICCP or Galvanic</td><td>13 mA/M^2</td><td>4 mA/M^2</td><td>65 mA/M^2</td><td>20 mA/M^2</td></tr>
</table>

Notes:

(1) Any section of pipeline which will be subjected to trenching and backfilling or any section of pipeline to be laid in such area which is expected to complete self-burial may be considered as "buried". However, any incomplete self-burials shall always be considered as "Pipeline Laid on Seabed"

(2) Cathodic protection systems for pipelines or electrically isolated sections of a pipeline that require 2.0 amperes or less may use galvanic anodes.

(3) All HDD installed pipeline shall be coated with CSD approved ARO coating system and CP system shall be designed for effective current distribution throughout the length of pipeline. Any HDD sections shall require a suitable attenuation calculations to verify protection can be achieved at either end AND the middle of the HDD. Also refer to section 6.2.8

Table 4 – Design Parameters for Galvanic Anode Materials

Anode Material	Consumption Rate (kg/A-Y) (2)	Open Circuit Potential (-mV) (1)
Aluminum	3.7	1050
Magnesium	10.3	1650
Zinc	11.8	1050

Note:

(1) Measured with reference to a silver/silver chloride reference electrode.

(2) Electrochemical capacity and open circuit potential values of the chosen pipeline anode shall be corrected for higher temperatures and burial status using Table 5 of ISO 15589-2.

Table 5 - Design Parameters for Impressed Current Anode Materials

Anode Material	Consumption (kg/A-Y)	Maximum Current Density (mA/cm^2)	Maximum Voltage (volt)
HSCI	0.45	0.7	No Limit
MMO	See Note 1	60.0	No Limit
Pt/Nb	8.63 x 10-6	40.0	60

Note:

(1) Use the consumption rate specified by the manufacturer of the MMO element, to determine the current output capacity and design life of MMO anodes.

Appendices

Appendix A - Coating & CP Requirements for Oil & Gas Fields Explained

Structures / CP Requirements	Electrified				Non - Electrified			
	Oil Fields		Gas Fields		Oil Fields		Gas Fields	
	With Well Casing	Without Well Casing	With Well Casing	Without Well Casing	With Well Casing	Without Well Casing	With Well Casing	Without Well Casing
Structure (Platform)	Bare		Coated	Bare (or) Coated	Shall be coated			
Well Casing	Bare	NA	Coated up to 10000[1] ft.	NA	Shall have coated well casings up to 4000[1] ft.	NA	Shall have coated well casings up to 10000[1] ft.	NA
CP Requirements	ICCP only	Shall be ICCP for bare ICCP or Galvanic (if coated)	ICCP only	Shall be ICCP for bare ICCP or Galvanic for coated	Galvanic CP			

Notes:

(1) Coating lengths are obtained from previous experience based on CPET logging results, below the specified lengths coating will not be effective due to high temperature.

Document History

07 July 2024	Major	Major revisions to incorporate references to several international standards, simulation modelling requirements, over protection limits, other items as listed in the revision under summary of changes.
08 December 2021	Editorial	Editorial revision to change the revision cycle from three to five years as per the updated SAEP-301.
02 May 2019	Editorial	Editorial revision to correct cross reference issue
11 March 2019	Major	Major revision to allow welded insert bracelet anode design with conditions and few minor changes
24 January 2018	Major	Major revision to incorporate VE Session ideas, committee member's comments, and to align with updated related company and industry standards for improvement
11 December 2013	Editorial	Editorial revision to paragraph 5.3.2.
19 September 2013	Editorial	Editorial changes to clarify bonding and negative cable requirements.
27 February 2013	Major	Revised the "Next Planned Update." Reaffirmed the content of the document, and made several editorial changes: Added the requirement for NACE certification for design engineers and field technicians Clarified the resistivity of seawater as 17 ohm-cm Added a section clarifying calculation requirements for circuit resistance Added a section clarifying power supply requirements Added a section to clarify remote monitoring requirements.